

WASP-12b

R-2301

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_150102 · created 2026-08-17T23:05:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457024.70029 +/- 0.00012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.045
Transit depth (Rp/Rs)^2	2.26 +/- 1.34 [%]
Orbital Inclination (inc)	86.71 +/- 2.96
Airmass coefficient 1 (a1)	2.45 +/- 1.41
Airmass coefficient 2 (a2)	-0.83 +/- 0.81
Scatter in the residuals of the lightcurve fit is	60.56 %
Best Comparison Star	None
Optimal Aperture	3.29
Optimal Annulus	31.79
Transit Duration (day)	0.1321 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.15 ± 0.045 vs 0.1178 (deviation 0.63σ)
Duration fit vs published	0.1321 d vs 0.125 d (deviation 1.09σ)
Mid-time O–C	-0.1 min
Depth SNR	3.0
Residual scatter	60.56 %

Light curve

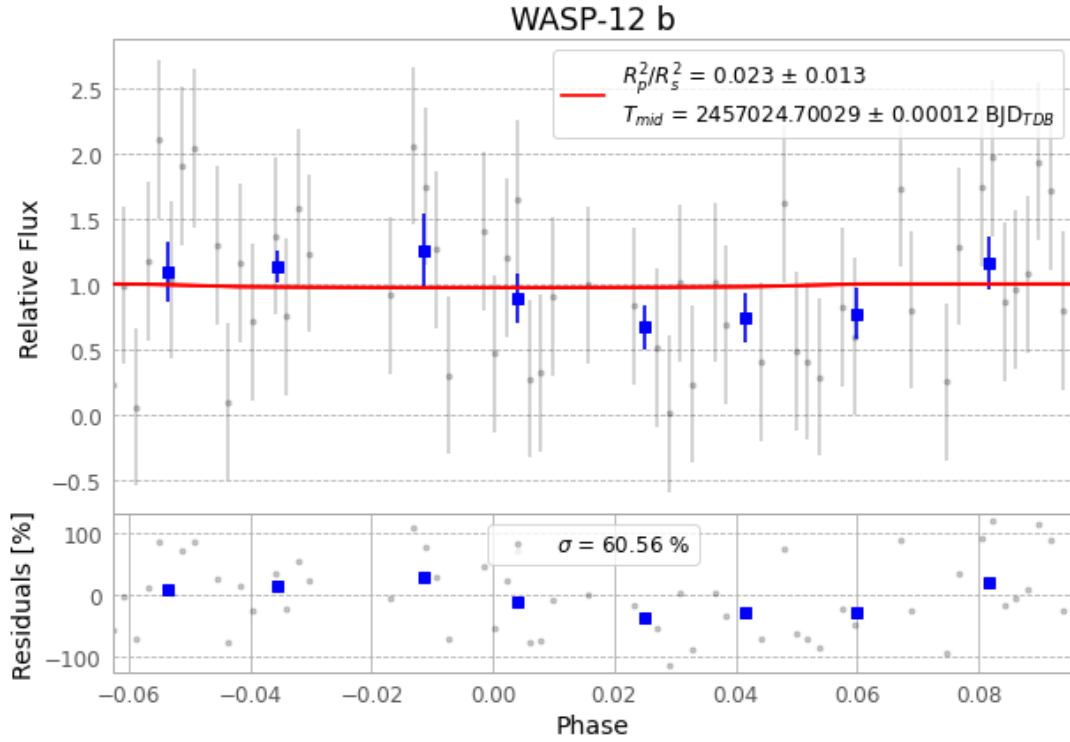


Figure 1 — FinalLightCurve_WASP-12 b_2015-01-01.png (SHA-256 74e9ea6e1f3453b9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [340, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 307172cebb8b386b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

84 FITS frames (55 MB), WASP-12150102030012.FITS → WASP-12150102070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-150102.log (30818d272a78...), FinalLightCurve_WASP-12 b_2015-01-01.png (74e9ea6e1f34...), FinalLightCurve_WASP-12 b_2015-01-01.csv (5123a31958fa...)

Compute resources

Wall time 246.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 23:05Z.